



Sydney Girls High School

2019

YEAR 11

Chemistry

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Student Number

Booklet 1 of 2

Part A & B

General Instructions

- Reading time – 5 minutes
- Working time – 60 minutes
- Write using black pen
- Draw diagrams using pencil
- Board-approved calculators may be used
- Two pages of data sheets and a Periodic Table are provided
- Write your student number on the multiple choice answer sheet and on each booklet

Total marks – 35

Part A – Pages 2-5

10 marks

- Attempt Questions 1 – 10

Part B – Pages 6-8

13 marks

- Attempt Questions 11 – 14

Part C – Pages 9 – 11

12 marks

- Attempt Questions 15 – 16

Part A – 10 marks**Attempt Questions 1 – 10**

Use the multiple-choice answer sheet for Questions 1–10

Question 1 and 2 refer to the following information.

A group of students conducted an experiment to investigate the reactivity of four different transition metals (A, B, C and D). They placed small pieces of each metal in separate test tubes with the appropriate test solutions. The result are summarised in the table below.

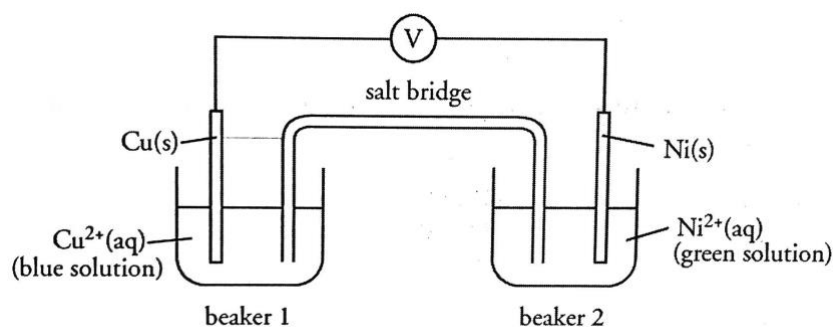
Metal ions Metal	$A^{3+}(aq)$	$B^{2+}(aq)$	$C^{4+}(aq)$	$D^{3+}(aq)$
A		No reaction	No reaction	Reaction occurs
B	Reaction occurs		No reaction	Reaction occurs
C	Reaction occurs	Reaction occurs		Reaction occurs
D	No reaction	No reaction	No reaction	

- Which of these metals is the most easily oxidised?

(A) Metal A
(B) Metal B
(C) Metal C
(D) Metal D
- Which of these metals is the weakest reducing agent?

(A) Metal A
(B) Metal B
(C) Metal C
(D) Metal D

- 3 Which half equation best represents the change when sodium metal reacts with dilute hydrochloric acid?
- (A) $\text{Na}^{2+} + 2\text{e}^- \rightarrow \text{Na(s)}$
- (B) $\text{Na}^{2+} \rightarrow \text{Na(s)} + 2\text{e}^-$
- (C) $\text{Na(s)} \rightarrow \text{Na}^+ + \text{e}^-$
- (D) $\text{Na(s)} + \text{e}^- \rightarrow \text{Na}^+$
- 4 The diagram below shows an electrochemical cell which has copper and nickel electrodes.



What three observations for this cell are correct?

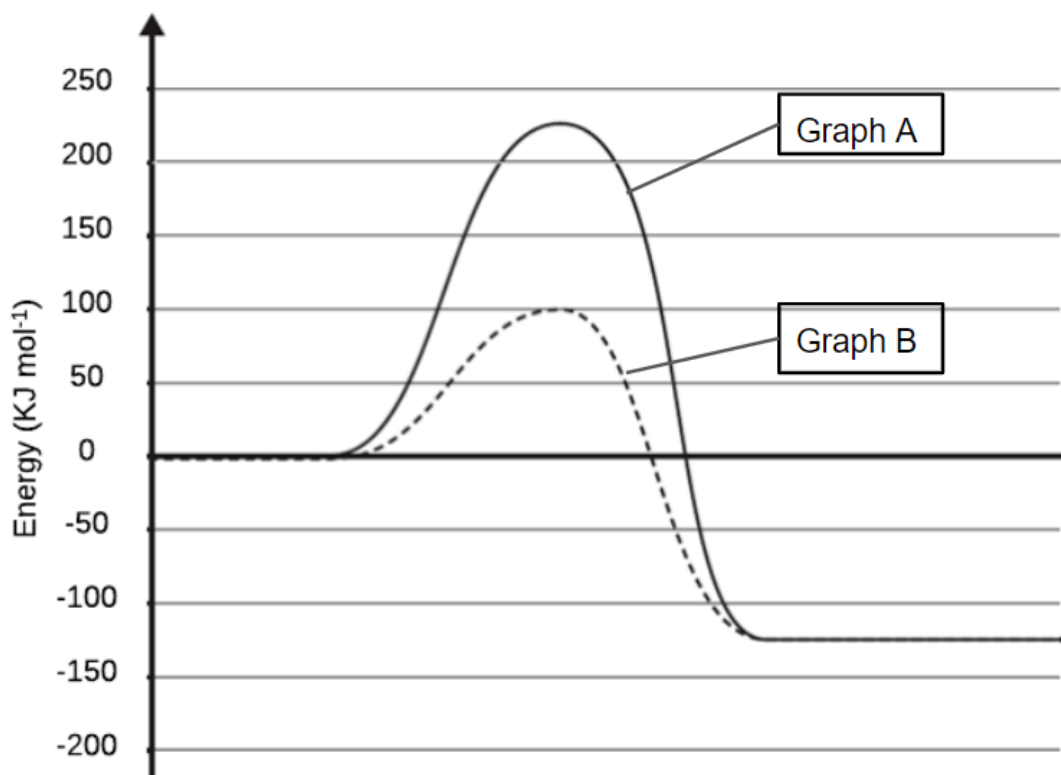
	Observation 1	Observation 2	Observation 3
(A)	The reading shown on the voltmeter was -0.57 V.	The solution in beaker 1 became a darker blue.	The solution in beaker 2 became a fainter green.
(B)	The reading shown on the voltmeter was -0.11 V.	A red/brown solid form on the copper electrode.	The solution in beaker 1 became a fainter blue.
(C)	A grey solid formed on the nickel electrode.	In beaker 1 the solution became a darker blue.	The solution in beaker 2 became a darker green.
(D)	The nickel electrode was the negative terminal.	A red/brown solid form on the copper electrode.	The solution in beaker 2 became a darker green.

- 5 The transition metal manganese exists in a range of different oxidation states.
- The oxidation states of Mn in MnO_2 and MnO_4^- respectively are,

- (A) +2 and +7
- (B) +3 and +6
- (C) +4 and +7
- (D) +6 and +4

Question 6 and 7 refer to the following information.

The energy profile diagram of a particular reaction is shown below. Graph A represent the uncatalysed reaction and Graph B represent the catalysed reaction.



6 Which of the following is correct?

	E_a of catalysed reaction (kJ mol ⁻¹)	ΔH of uncatalysed reaction (kJ mol ⁻¹)
(A)	100	-125
(B)	100	-350
(C)	225	-125
(D)	225	-350

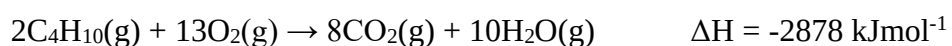
7 What would be the activation energy of the reverse of the reaction in Graph A?

- (A) 100 kJ mol⁻¹
- (B) 225 kJ mol⁻¹
- (C) 250 kJ mol⁻¹
- (D) 350 kJ mol⁻¹

- 8 When hydrochloric acid reacts with potassium hydroxide, energy is released because
- (A) the reaction involves the breaking of bonds.
 - (B) the stored energy in the products is greater than the stored energy in the reactants.
 - (C) the activation energy is greater than the energy released from the hydrogen and oxygen molecules.
 - (D) as the bonds in the products form, more energy is released than is required to break the bonds in the reactants.
- 9 Under what circumstances will a change be nonspontaneous regardless of the temperature?

	ΔH	ΔS	$-T\Delta S$	ΔG
(A)	+	–	+	+
(B)	–	+	–	–
(C)	–	–	+	+
(D)	+	+	–	–

- 10 The equation for the reaction of butane with oxygen is



When 22g of carbon dioxide is produced in this reaction, the energy change will be:

- (A) 180 kJ released
- (B) 360 kJ released
- (C) 1439 kJ released
- (D) 2878 kJ released

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Student Number

Part B – 13 marks

Attempt Questions 11 - 14

Answer the questions in the spaces provided.

Show all relevant working in questions involving calculations.

Question 11 (4 marks)

The formulae of 10 substances are given below:

NaOH	KCl	AgNO ₃	CaCO ₃	N ₂
C ₂ H ₆	O ₂	H ₂ SO ₄	H ₂	CO ₂

- a) Identify TWO substances that would react together in an aqueous environment to form a precipitate:

- i) Write a balanced overall chemical equation including states. 1K

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- ii) Write a net ionic equation including states. 1K

.....

- b) Which TWO substances will take part in an acid and base reaction in an aqueous environment?

- i) Write a balanced overall chemical equation including states. 1K

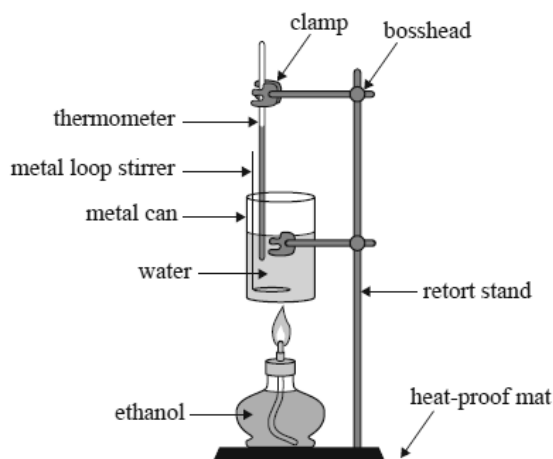
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- ii) Write a net ionic equation including states. 1K

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Question 12 (4 marks)

Students were asked to perform a first-hand investigation to determine the molar heat of combustion of ethanol ($\text{C}_2\text{H}_5\text{OH}$). The following extract is from the practical report of one student.



Lab data:

Mass of water	= 150.0 g
Initial mass of burner	= 182.5 g
Final mass of burner	= 181.1 g
Initial temperature of water	= 24.0°C
Final temperature of water	= 64.0°C

After completing the calculations correctly, the student found that the answer did not agree with the value found on data sheets.

- a) Suggest TWO reasons for this. 2P

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- b) Propose TWO ways the student can improve the accuracy of the results. 2P

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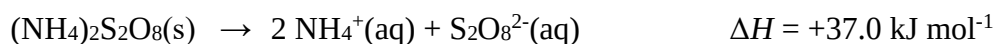
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Question 13 (3 marks)

The compound ammonium peroxydisulfate, $(\text{NH}_4)_2\text{S}_2\text{O}_8$, dissolves in water according to the following equation.



In an experiment, 22.82 g of ammonium peroxydisulfate was dissolved in 100 g of water. If the final temperature of the water was 15.0°C , what is the initial temperature? 3A

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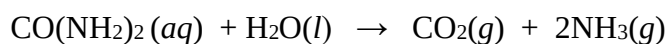
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Question 14 (2 marks)

The following equation shows the reaction of urea with water at 25°C ,



The enthalpy change ΔH° for this reaction is $+119.2 \text{ kJ mol}^{-1}$ and the entropy change ΔS° is $+354.8 \text{ J mol}^{-1} \text{ K}^{-1}$. Calculate the Gibbs free energy change, then predict the spontaneity of this reaction. Show all working. 2A

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END OF PART B

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Part C – 12 marks**Attempt Questions 15 - 16**

Answer the questions in the spaces provided.

Show all relevant working in questions involving calculations.

Question 15 (10 marks)

- a) You conducted an experiment to determine the effect of concentration on rate of reaction. Describe a method to safely investigate this, including a relevant chemical equation.

5P

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Question 15 - continued

- b) With the aid of a diagram, explain the effect of increasing the concentration of reactants on the rate of reaction. 3K

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- c) Explain the effect of increasing temperature on the rate of reaction. 2K

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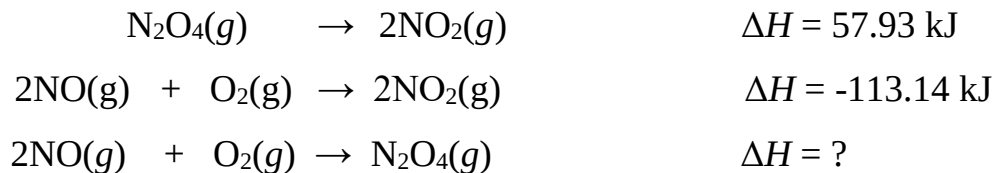
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END OF PART C



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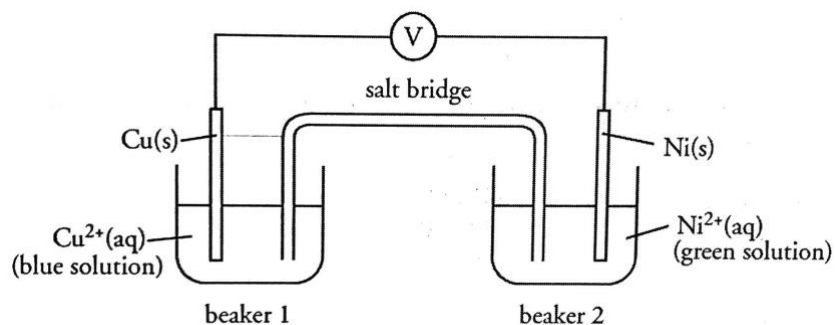
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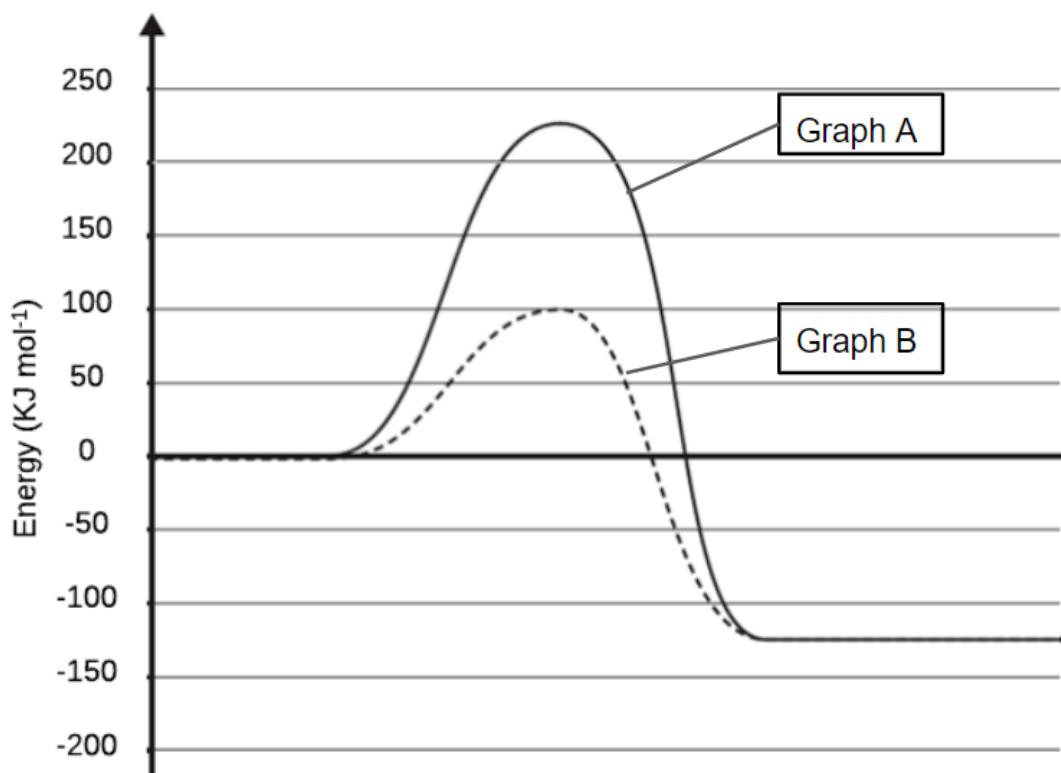
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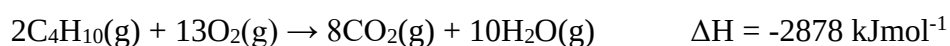
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1	C	2	D
3	C	4	D
5	C	6	A
7	D	8	D
9	A	10	A

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Part B – 13 marks

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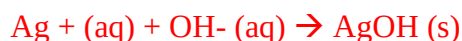
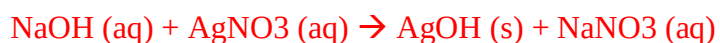
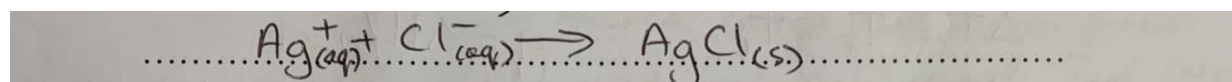
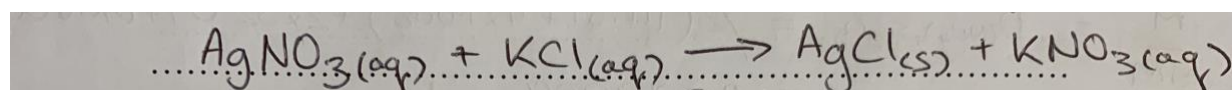
- a) Identify TWO substances that would react together in an aqueous environment to form a precipitate:

i) Write a balanced overall chemical equation including states. 1K

ii) Write a net ionic equation including states.

1K

Correct balanced chemical equation with correct states	1
Correct net ionic equation with correct states	1



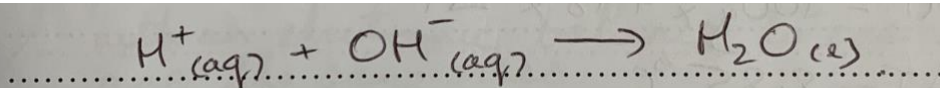
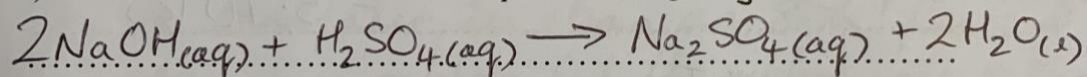
- calcium carbonate has very low solubility 0.013 g/L at 25°C

b) Which TWO substances will take part in an acid and base reaction in an aqueous environment?

i) Write a balanced overall chemical equation including states. 1K

ii) Write a net ionic equation including states. 1K

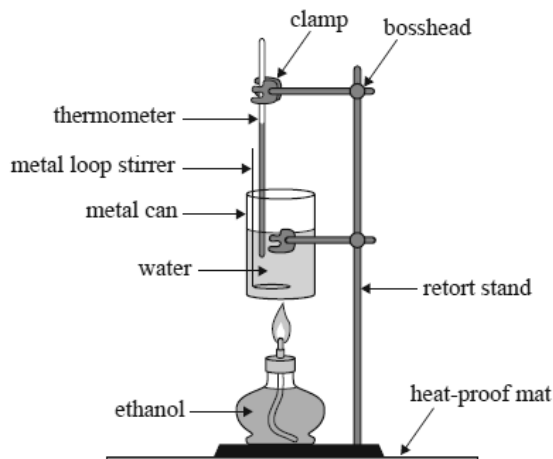
Correct balanced chemical equation with correct states	1
Correct net ionic equation with correct states	1



- Simplify $2\text{H}^+_{(aq)} + \text{OH}^-_{(aq)} \rightarrow \text{H}_2\text{O}_{(l)}$

Question 12 (4 marks)

Students were asked to perform a first-hand investigation to determine the molar heat of combustion of ethanol ($\text{C}_2\text{H}_5\text{OH}$). The following extract is from the practical report of one student.



Lab data:

Mass of water	= 150.0 g
Initial mass of burner	= 182.5 g
Final mass of burner	= 181.1 g
Initial temperature of water	= 24.0°C
Final temperature of water	= 64.0°C

After completing the calculations correctly, the student found that the answer did not agree with the value found on data sheets.

a) Suggest TWO reasons for this.

2P

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Two correct reasons provided • Heat loss to surroundings • Standard conditions (298K and 1 atm) might not apply • Incomplete combustion	2
One Correct reason provided	1

b) Propose TWO ways the student can improve the accuracy of the results.

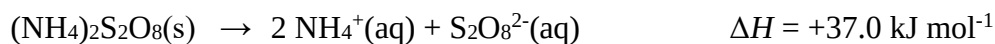
2P

Two correct improvements provided e.g. • keep the distance between the wick and the calorimeter as low as possible • a lid or insulation could be used to reduce heat loss of the calorimeter • ensure the thermometer is not touching the bottom of the calorimeter	2
One Correct improvement provided	1

- Using more precise equipment will not improve the reasons provided in a) significantly

Question 13 (3 marks)

The compound ammonium peroxydisulfate, $(\text{NH}_4)_2\text{S}_2\text{O}_8$, dissolves in water according to the following equation.



In an experiment, 22.82 g of ammonium peroxydisulfate was dissolved in 100 g of water. If the final temperature of the water was 15.0°C , what is the initial temperature? 3A

• The number of mole for ammonium peroxydisulfate is correctly calculated $n = 0.09999 \text{ mol}$	1
• Correctly find $q = \Delta H \times n = 3.70 \text{ kJ}$	1
• Correct substitution into $q = mc\Delta T$ and find $T = 23.9^\circ\text{C}$	1

$\frac{m}{MM} = \dots n_{(\text{NH}_4)_2\text{S}_2\text{O}_8} = \frac{22.82}{284.10} = 0.09999 \text{ mol}$

$\Delta T = T_f - T_i$ $q = \Delta H \times n$

$\dots = 37.0 \text{ kJ mol}^{-1} \times 0.1 \text{ mol}$

$\dots = 3.70 \text{ kJ}$

$q = mc\Delta T$

$3.70 = 100 \text{ g} \times 4.18 \times \Delta T$

$\Delta T = -8.85^\circ\text{C}$

$T_i = T_f - \Delta T$

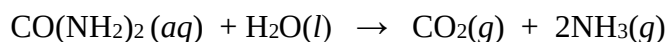
$\dots = 15.0 - (-8.85)$

$\dots = 23.85$

$\dots = 24^\circ\text{C} \text{ (to 2 sig figs)}$

Question 14 (2 marks)

The following equation shows the reaction of urea with water at 25 °C,



The enthalpy change ΔH° for this reaction is +119.2 kJ mol⁻¹ and the entropy change ΔS° is +354.8 J mol⁻¹ K⁻¹. Calculate the Gibbs free energy change, then predict the spontaneity of this reaction. Show all working.

2A

• Correct substitution into $\Delta G = \Delta H - T\Delta S$, and find $\Delta G = +13.5 \text{ kJ mol}^{-1}$	1
• Correct prediction from the sign of ΔG	1

$$\begin{aligned}\Delta G &= \Delta H - T\Delta S \\ &= +119.2 \text{ kJ mol}^{-1} - 298 \times 354.8 \text{ J mol}^{-1} \text{ K}^{-1} \\ &= +119.2 \times 10^3 \quad 105730 \\ &= 119200 \text{ J mol}^{-1} - 105730 \text{ J mol}^{-1} \\ &= +13.5 \text{ kJ mol}^{-1}.\end{aligned}$$

As ΔG is +ve, reaction will not be spontaneous at this temperature.

END OF PART B

Booklet 2 of 2

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Student Number

Part C – 12 marks

Attempt Questions 15 - 16

Answer the questions in the spaces provided.

Show all relevant working in questions involving calculations.

Question 15 (10 marks)

- a) You conducted an experiment to determine the effect of concentration on rate of reaction. Describe a method to safely investigate this, including a relevant chemical equation.

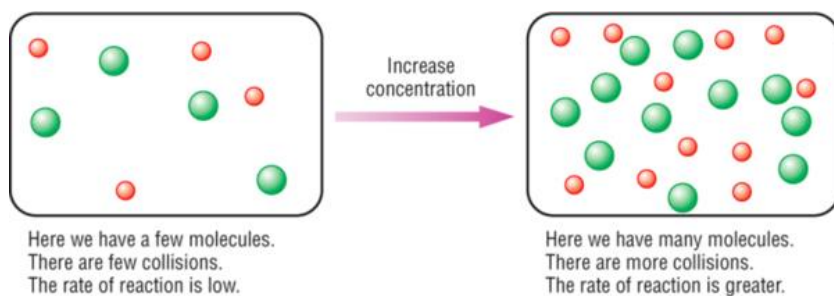
5P

- Consider the safety aspect of the experiment - Experiment must be fair, controlled, observable and repeatable - Correct independent and dependent variables identified - All equipment used must be identified (stopwatch, electronic balance) - One relevant chemical equation included	5
Missing one or more points	2-4
Some relevant information	1

- b) With the aid of a diagram, explain the effect of increasing the concentration of reactants on the rate of reaction.

3K

- Increasing concentration of reactants increases rate of reaction - Increasing the concentration means there are more reactant particle/ increases the frequency of collisions between reactants in a given space (volume). This increases the chance that reactant particles will collide. The increased frequency of collisions results in a faster rate of reaction. - Relevant labelled diagram provided	3
Missing one point as indicate above	2
One correct point provided	1



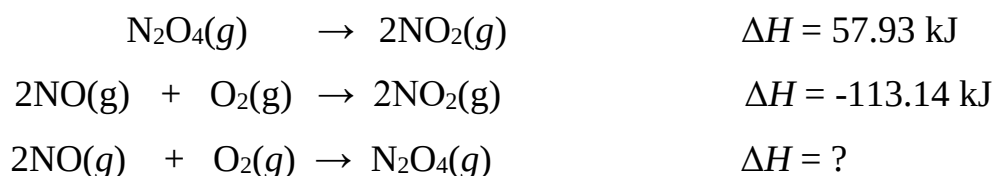
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2K

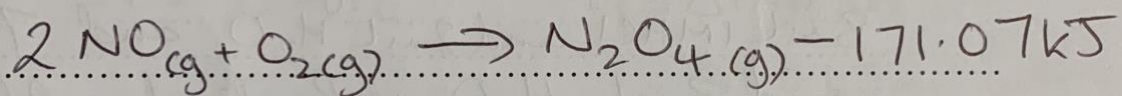
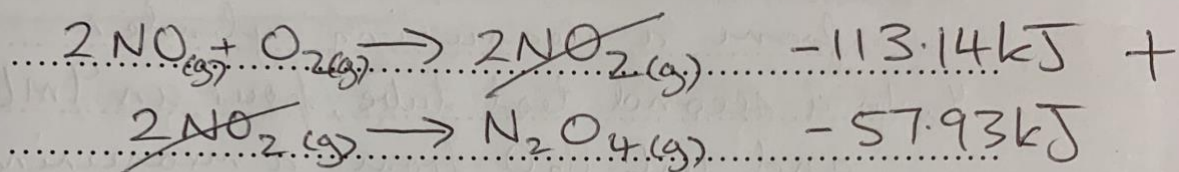
- Increasing temperature increases rate of reaction - As the temperature increases 1. the particles move faster resulting in more collisions which in turn leads to a faster reaction. And 2. There are more collisions which have enough activation energy to result in a successful collision.	2
One correct reason provided	1

Question 16 (2 marks)

Use the following equations to calculate the enthalpy of dinitrogen tetroxide formation. 2A
Show all working.



Correctly rearrange the first equation $2\text{NO}_2(g) \rightarrow \text{N}_2\text{O}_4(g) \quad \Delta H = -57.93 \text{ kJ}$	1
Correctly adding the equation together and find $\Delta H = -171.07 \text{ kJ}$, with the correct unit	1
* Lose 1 mark if there is no correct unit provided	



END OF PART C